

Krishna vamsi Dhulipalla

Blacksburg, Virginia

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Skills

Programming & Web:	Python, JavaScript, TypeScript, ReactJS, NodeJS, Flask, REST APIs, MongoDB, MySQL, AngularJS, Django, Bootstrap, FastAPI, HTML5, CSS3, Express.js, Firebase
Machine Learning/AI:	PyTorch, Tensorflow, Supervised and Unsupervised Learning, Transformers, NER, Sentiment Analysis, Transfer Learning, Self-Supervised Learning, Federated Learning, Scikit-Learn, YOLO, Computer Vision (ResNet, VGG, GAN, U-Net), NLP, Reverse Gradient, Domain Adaptation, Generative Models, Fine-Tuning LLMs, Cross-Domain Techniques, SHAP analysis
Data Analysis:	Pandas, Tableau, D3.js, Numpy, Matplotlib, Seaborn, Scipy
DevOps & Cloud:	AWS(S3, Lambda, EMR, Glue, ECS, SQS), Docker, Git, Azure, Apache Spark, ETL pipelines, Apache Kafka, Apache Airflow, Kubernetes

Experience

Research Assistant - ML in Genomics

Sep. 2024 - Present

Virginia Tech

Blacksburg, VA

- Conducted research on LLMs for bioinformatics, focusing on inference, fine-tuning, and model adaptation for genomics and plant species-specific DAP-seq data.
- Experimented with DNA-specialized models like DNABERT and HyenaDNA, customizing model architecture and tokenization approaches to handle genomic sequences, k-mer, and DAP peaks effectively.
- Engineered preprocessing pipelines for over 100,000 sequences, applying feature extraction methods to optimize data representation and enhance model performance.

Graduate Research Assistant

May. 2023 - June. 2024

Virginia Tech

Blacksburg, VA

- Advanced cross-domain Named Entity Recognition using NLP techniques, which improves adaptation across domains without changing pre-trained weights.
- Automated pipelines were developed to reduce manual intervention by 25% and optimize the end-to-end data processing for NER jobs.
- Optimized NLP workflows for scalability, improving processing time by 30% for large, heterogeneous datasets.

Data Engineer

Jan. 2022 - Nov. 2022

HCL

Hyderabad, India

- Developed and optimized automated ETL pipelines using Python to extract, clean, and transform large datasets from eCommerce websites, resulting in a 20% increase in data processing efficiency.
- Transitioned from batch processing to real-time data processing by implementing Apache Kafka for streaming data and Apache Spark for rapid analysis, reducing data latency by 30%
- Containerized the entire pipeline, including Kafka and Spark components, with Docker, ensuring consistent performance across various environments and improving deployment speed by 25%

Projects

Cross-Domain Classification of Gene Rhythmicity
Using DNABERT and k-mer Tokenization

- Generated embeddings using the DNABERT model, applying tokenization with different k-mer sizes (3-mers, 4-mers, 6-mers, and 8-mers) to process DNA sequences effectively.
- Used the DNABERT model to classify rhythmicity classes on mRNAs gene data, leveraging advanced tokenization and embedding techniques to capture sequence features.
- Tested the model on lncRNA sequences without training on lncRNA (cross-domain evaluation) to assess the model’s ability to generalize across different types of gene data.
- **keywords :** DNABERT, mRNA, incRNA, Cross-domain

Proxy TuNER: Advancing Cross-Domain Named
Entity Recognition through Proxy Tuning

[Source Link](#)

- Applied proxy tuning to BERT for cross-domain NER, using an LSTM to adjust logits and improve domain adaptability without modifying pre-trained weights.
- Implemented reverse gradient for domain generalization, boosting performance by learning domain-invariant features.
- Reduced computational costs by 25% and increased processing efficiency by 35%, improving scalability and adaptation to new domains.
- Achieved an average F1-score of 73% across domains, with peak performance of 78%.
- **keywords:** Cross-Domain NER, Proxy Tuning, Domain Generalization, Reverse Gradient

IntelliMeet: AI-Enabled Decentralized Video
Conferencing App

[Source Link](#)

- Developed an AI-enabled decentralized video conferencing app, supporting over 10 concurrent users with real-time video processing.
- Implemented federated learning, enabling local model training on user devices and sending gradients to a central server, which aggregated updates into a global model, improving accuracy by 25%.
- Achieved 83% speech-to-text accuracy, enhancing user engagement by 25% and reducing latency by 15%, significantly improving overall user experience.
- **keywords :** Decentralized Video Conferencing, WebRTC, Federated Learning

Publications

- Miao, L.,Dhulipalla, K ., Kundu, S ., Manthapuri, V., & Li, S. (2024). Leveraging Machine Learning for Predicting Circadian Transcription in mRNAs and lncRNAs (*Submitted to IEEE BIBM 2024 on Sept 23 2024*)

Education

Virginia Tech

M.S. in Computer Science CGPA - 3.95/4

Blacksburg, Virginia

Jan. 2023 - Dec. 2024

Vel Tech University

B.Tech in Computer Science and Engineering CGPA - 8.24/10

Chennai, Inida

Jun. 2018 - May. 2022

Certifications

- Amazon AWS (Cloud Solutions) →
- Cisco Cyber Ops (Digital Security) →
- Microsoft MERN Development (Full-Stack Development) →
- Coursera Machine Learning (Data-Driven Insights, ML Algorithms) →
- Google Cloud Data Engineering (Scalable Data Pipelines, BigQuery, Data Lakes) →